

SRS and Feature List

Functional Requirements

Checkout

FR-001 The system shall require each user to sign in under an authorized role, including Cashier, Manager, Administrator, or Head-Office Manager.

FR-002 The system shall allow an authorized cashier to start a new sale.

FR-003 The system shall allow the cashier to add products, change quantities, and remove products before the sale is completed.

FR-004 The system shall display and update the subtotal, discount, VAT information, and final amount whenever the contents of the sale change.

FR-005 The system shall provide one Discount Percentage field and one Discount Amount field for an order-level discount.

FR-006 The system shall allow the cashier to cancel a sale at any time before completion.

Payment

FR-007 The system shall allow the cashier to select either cash or credit card as the payment method.

FR-008 For a cash payment, the system shall accept the amount tendered and calculate the change due.

FR-009 Before requesting card authorization, the system shall assign a unique reference to the card-payment attempt and record the pending payment attempt durably.

FR-010 If the system cannot record the pending card-payment attempt, it shall not send a payment request to the card machine or Payment Service Provider.

FR-011 For a credit-card payment, the system shall send the final sale amount and unique payment-attempt reference to the connected card machine and Payment Service Provider.

FR-012 The system shall receive and display the approval or decline result returned by the Payment Service Provider.

FR-013 When a card payment is declined, the system shall allow the cashier to retry the card payment, change the payment method to cash, or cancel the sale.

FR-014 When the Payment Service Provider is unavailable before a card-payment request is sent, the system shall inform the cashier that card payment is unavailable and allow the cashier to accept cash or cancel the sale.

FR-015 If communication is interrupted after a card-payment request has been sent, the system shall treat the payment result as unresolved and query the Payment Service Provider using the original payment-attempt reference.

FR-016 While a card-payment result remains unresolved, the system shall prevent the cashier from initiating another payment, changing to cash, or cancelling the sale.

FR-017 If card payment is approved but the system cannot finalize the sale, the system shall retain the payment attempt for recovery and shall not initiate another card charge.

FR-018 The system shall automatically reconcile unresolved card-payment attempts with the Payment Service Provider using their original payment-attempt references.

FR-019 If reconciliation confirms that a card payment was approved, the system shall finalize the sale without submitting another card charge.

FR-020 If reconciliation confirms that a card payment was declined or was not processed, the system shall return the sale to an unpaid state and allow the cashier to retry the card payment, change to cash, or cancel the sale.

FR-021 If the Payment Service Provider cannot determine the payment result, the system shall retain the payment attempt in a recovery-pending state and prevent another payment attempt until the result is resolved.

FR-022 The system shall allow authorized personnel to view card-payment attempts that remain unresolved or require recovery.

Sale Recording and Receipt

FR-023 The system shall record each completed sale with its sale identifier, store, terminal, cashier, date and time, sold items, discount, VAT, final amount, payment method, and applicable payment reference.

FR-024 The system shall print a receipt using the store receipt printer after a sale is completed successfully.

FR-025 If receipt printing fails, the system shall inform the cashier and allow an authorized user to reprint the receipt without reversing or duplicating the completed sale.

Inventory Integration

FR-026 When finalizing a sale, the system shall durably record the completed sale and its pending inventory-update record as one atomic operation.

FR-027 If the atomic operation cannot be completed, the system shall not mark the sale as completed. The system shall keep a cash sale open and inform the cashier, or place an already-approved card payment into the payment-recovery process.

FR-028 Each pending inventory-update record shall contain a unique sale identifier, store identifier, terminal identifier, transaction date and time, and the product and quantity of each item sold.

FR-029 The system shall asynchronously send each pending inventory-update record to the inventory messaging service.

FR-030 If an inventory-update record cannot be delivered, the system shall retain it and retry delivery until the inventory messaging service acknowledges receipt.

FR-031 Failure to deliver an already stored inventory-update record shall not reverse or block its corresponding completed sale.

FR-032 The system shall allow authorized personnel to view inventory-update records that are awaiting delivery or have repeatedly failed delivery.

Daily Sales Summary

FR-033 The system shall allow a Manager to view the daily sales summary for the Manager's assigned store and a selected date.

FR-034 The system shall allow an Administrator or Head-Office Manager to view the daily sales summary for any selected store.

FR-035 The system shall allow an Administrator or Head-Office Manager to view a consolidated daily sales summary across all stores.

FR-036 Each daily sales summary shall show the total amount received by cash and the total amount received by credit card separately.

Product and Price Management

FR-037 The system shall allow an authorized Manager to add a product to the product catalog used by the POS.

FR-038 The system shall allow an authorized Manager to update product information and selling prices within the Manager's permitted scope.

Administration

FR-039 The system shall allow an Administrator to create and deactivate user accounts.

FR-040 The system shall allow an Administrator to assign roles and permitted stores to users.

FR-041 The system shall allow an Administrator to create, update, and deactivate stores.

FR-042 The system shall allow an Administrator to register, activate, and deactivate POS terminals for a store.

Product Identification at Checkout

FR-043 The system shall allow a cashier to add a product by barcode scan, manual barcode entry, or search by product code or name.

2. Business Rules

BR-001 All product selling prices shall be expressed in Thai baht and shall include VAT according to applicable Thai VAT rules.

BR-002 The sale subtotal shall equal the sum of the extended amounts of all sale items before applying the order-level discount.

BR-003 A sale may have only one order-level discount. Product-line discounts and multiple combined discounts are not supported.

BR-004 When a user enters or changes the Discount Percentage, the system shall calculate the Discount Amount as:

$$\text{Discount Amount} = \text{Sale Subtotal} \times \text{Discount Percentage} \div 100$$

BR-005 A percentage-based Discount Amount shall be rounded to the nearest satang and shall be the amount used to calculate the final sale total.

BR-006 When a user enters or changes the Discount Amount directly, the system shall set the Discount Percentage to null and use the entered amount as the discount.

BR-007 When the sale contents change while a Discount Percentage is active, the system shall recalculate the Discount Amount using the revised subtotal.

BR-008 The Discount Percentage shall be between 0 and 100 percent, inclusive.

BR-009 The Discount Amount shall not be negative and shall not exceed the sale subtotal.

BR-010 The final sale amount shall equal the subtotal minus the Discount Amount.

BR-011 A sale shall use exactly one payment method. Split payment between cash and credit card is not permitted.

BR-012 A cash sale shall not be completed unless the amount tendered is equal to or greater than the final sale amount.

BR-013 A credit-card sale shall not be completed unless the Payment Service Provider approves payment for the final sale amount.

BR-014 A sale shall be considered completed after successful payment and permanent recording of the sale. Receipt-printing failure shall not reverse a completed sale.

BR-015 A cancelled sale shall not be included in sales totals and shall not produce an inventory message.

BR-016 The physical presence of an item at the checkout counter shall be treated as sufficient evidence that it may be sold. The POS shall not synchronously check or block a sale based on the recorded inventory balance.

BR-017 Daily cash and card totals shall include completed sales for the selected store or stores and selected local calendar date, after discounts.

BR-018 A Manager may view daily summaries only for the manager's assigned store.

BR-019 An Administrator or Head-Office Manager may view summaries for any individual store or for all stores combined.

BR-020 Returns, refunds, and refund receipts shall not be processed by the POS.

BR-021 The product catalog and selling price of each product shall be global and shall apply to all stores.

3. Non-Functional Requirements

NFR-001 Performance: Under normal operating conditions, the system shall respond to checkout actions, excluding external payment processing and physical printing, within two seconds for at least 95 percent of requests.

NFR-002 Reliability: A successfully recorded sale shall not be lost because the receipt printer, Payment Service Provider, messaging service, or Inventory Service becomes unavailable afterward.

NFR-003 Messaging Reliability: Inventory messages shall be stored durably and delivered using a mechanism that supports retry without losing completed sales.

NFR-004 Duplicate Prevention: Each inventory message shall contain a unique transaction identifier so that repeated delivery does not cause the Inventory Service to deduct the same stock more than once.

NFR-005 Payment Continuity: Failure of the Payment Service Provider shall not prevent the POS from completing an otherwise valid cash sale.

NFR-006 Security: The system shall enforce role-based access to checkout, reporting, and product-management functions.

NFR-007 Card Security: The POS shall not store sensitive card details and shall communicate with the card machine and Payment Service Provider through secure, approved interfaces.

NFR-008 Auditability: The system shall record the responsible user, date, time, store, and terminal for completed sales, sale cancellations, discounts, and product or price changes.

NFR-009 Scalability: The system shall support multiple stores and multiple concurrently operating POS terminals without transaction-identifier collisions.

NFR-010 Localization: The system shall represent monetary values in Thai baht to two decimal places and use the store's local date and time for daily reporting.

NFR-011 Interoperability: The system shall integrate with the approved receipt printer, card machine, Payment Service Provider, and asynchronous inventory-messaging interface.

NFR-012 Online Connectivity: A terminal shall require connectivity to the Kaching backend to start or complete a sale; offline sales are not supported.

NFR-013 Capacity and Throughput: The system shall support 20 stores, 10 active terminals per store, 200 concurrent users, and at least 20 completed sales per second.

NFR-014 Availability: The Kaching backend shall provide 99.5 percent monthly availability.

NFR-015 Disaster Recovery: The production service shall meet an RPO of 15 minutes and an RTO of 4 hours.

NFR-016 Usability and Accessibility: The English-only user interface shall support desktop and touchscreen use at 1280 x 720 or higher and meet WCAG 2.1 AA color contrast.

Complete Feature Inventory: POS System

Application Features

ID	Feature	Feature scope
Feature-A	User Authentication and Access Control	Sign-in, user roles, function permissions, and store-based access restrictions for Cashiers, Managers, Administrators, and Head-Office Managers.
Feature-B	Store and Terminal Operating Context	Identification of the current store and POS terminal, association of transactions with that context, support for multiple stores and terminals, and enforcement of store and terminal activation status and valid terminal-to-store associations.
Feature-C	Product Catalog and Price Management	Addition and modification of products, product information, globally applicable VAT-inclusive selling prices, and manager access restrictions.
Feature-D	Sale Lifecycle Management	Starting a sale, maintaining its current state, cancelling it before completion, and preventing modification after completion.
Feature-E	Cart and Sale-Item Management	Adding products by barcode scan, manual barcode entry, or search by product code or name; changing quantities; removing products; and recalculating the sale when its items change.

Feature-F	VAT-Inclusive Pricing and Total Calculation	Calculation and display of item amounts, subtotal, Thai VAT information, discount, and final sale amount using Thai baht and satang.
Feature-G	Order-Level Discount Management	Discount Percentage and Discount Amount entry, percentage-to-amount calculation, satang rounding, manual discount-amount handling, validation, and recalculation when the cart changes.
Feature-H	Payment Method Selection	Selection of either cash or credit card, enforcement of the no-split-payment rule, and permitted switching of payment methods.
Feature-I	Cash Payment Processing	Entry of cash tendered, validation that sufficient cash was received, change calculation, and cash-sale completion.
Feature-J	Card Payment Authorization	Creation of a unique payment-attempt reference, durable recording of the attempt, communication with the card machine and Payment Service Provider, and handling of approved or declined results.
Feature-K	Card Payment Recovery and Reconciliation	Handling communication interruptions, unresolved payment results, approved payments that could not be finalized, PSP status queries, duplicate-charge prevention, and recovery of affected sales.
Feature-L	Sale Finalization and Durable Recording	Validation of successful payment, assignment of a unique sale identifier, permanent recording of the sale, and atomic creation of the associated pending inventory update.
Feature-M	Receipt Printing and Reprinting	Printing a receipt after successful sale completion, reporting printer failures, and reprinting without duplicating or reversing the sale.

Feature-N	Inventory Update Outbox	Creation and durable storage of a pending inventory-update record containing the sale, store, terminal, date and time, product, and quantity information.
Feature-O	Inventory Message Delivery	Asynchronous message transmission, acknowledgement processing, retained-message retry, and duplicate inventory-update prevention through unique transaction identifiers.
Feature-P	Operational Exception Monitoring	Viewing unresolved card payments, payment-recovery cases, pending inventory messages, and inventory messages that have repeatedly failed delivery.
Feature-Q	Daily Sales Reporting	Selection of a business date, assigned-store reporting for Managers, selected-store reporting for Administrators and Head-Office Managers, consolidated reporting, and separate cash and card totals.
Feature-R	Audit Trail	Recording the responsible user, store, terminal, date, and time for completed sales, cancellations, discounts, card-payment recovery, and product or price changes.
Feature-S	Administration and Configuration Management	Creation and deactivation of user accounts; assignment of roles and permitted stores; store creation, update, and deactivation; and POS terminal registration, activation, and deactivation.

Cross-Cutting Quality Constraints

The following NFR concerns apply across the relevant features and must be addressed during system-level and feature-level design:

1. **Performance:** Checkout operations must meet the specified response-time target under normal conditions.
2. **Reliability:** Completed sales and pending inventory updates must not be lost.
3. **Transaction consistency:** Sale completion and inventory-outbox creation must be atomic.

4. **Payment safety:** The system must prevent duplicate card charges and reconcile uncertain payment results.
5. **Messaging reliability:** Inventory updates must support durable storage, retry, acknowledgement, and idempotent processing.
6. **Security:** Role-based access must be enforced throughout the application.
7. **Card-data security:** Sensitive card information must not be stored by the POS, and approved secure interfaces must be used.
8. **Auditability:** Important financial and administrative actions must be traceable to the responsible user and terminal.
9. **Scalability and concurrency:** Multiple stores and POS terminals must operate concurrently without transaction-identifier collisions.
10. **Online connectivity:** A terminal must be connected to the Kaching backend to start or complete a sale; offline sales are not supported.
11. **Capacity and throughput:** The system must support the specified numbers of stores, active terminals, concurrent users, and completed sales per second.
12. **Availability:** The Kaching backend must meet the specified monthly availability target.
13. **Disaster recovery:** The production service must meet the specified recovery point and recovery time objectives.
14. **Usability and accessibility:** The English-only interface must support the specified desktop and touchscreen resolution and meet WCAG 2.1 AA color-contrast requirements.

POS Requirements-to-Features Traceability Matrix

1. Functional Requirements

Requirement	Requirement statement	Covered by feature(s)
FR-001	The system shall require each user to sign in under an authorized role, including Cashier, Manager, Administrator, or Head-Office Manager.	Feature-A: User Authentication and Access Control
FR-002	The system shall allow an authorized cashier to start a new sale.	Feature-A: User Authentication and Access Control; Feature-D: Sale Lifecycle Management
FR-003	The system shall allow the cashier to add products, change quantities, and remove products before the sale is completed.	Feature-E: Cart and Sale-Item Management

FR-004	The system shall display and update the subtotal, discount, VAT information, and final amount whenever the contents of the sale change.	Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-G: Order-Level Discount Management
FR-005	The system shall provide one Discount Percentage field and one Discount Amount field for an order-level discount.	Feature-G: Order-Level Discount Management
FR-006	The system shall allow the cashier to cancel a sale at any time before completion.	Feature-D: Sale Lifecycle Management
FR-007	The system shall allow the cashier to select either cash or credit card as the payment method.	Feature-H: Payment Method Selection
FR-008	For a cash payment, the system shall accept the amount tendered and calculate the change due.	Feature-I: Cash Payment Processing
FR-009	Before requesting card authorization, the system shall assign a unique reference to the card-payment attempt and record the pending payment attempt durably.	Feature-J: Card Payment Authorization
FR-010	If the system cannot record the pending card-payment attempt, it shall not send a payment request to the card machine or Payment Service Provider.	Feature-J: Card Payment Authorization

FR-011	For a credit-card payment, the system shall send the final sale amount and unique payment-attempt reference to the connected card machine and Payment Service Provider.	Feature-J: Card Payment Authorization
FR-012	The system shall receive and display the approval or decline result returned by the Payment Service Provider.	Feature-J: Card Payment Authorization
FR-013	When a card payment is declined, the system shall allow the cashier to retry the card payment, change the payment method to cash, or cancel the sale.	Feature-J: Card Payment Authorization; Feature-H: Payment Method Selection; Feature-D: Sale Lifecycle Management
FR-014	When the Payment Service Provider is unavailable before a card-payment request is sent, the system shall inform the cashier that card payment is unavailable and allow the cashier to accept cash or cancel the sale.	Feature-J: Card Payment Authorization; Feature-H: Payment Method Selection
FR-015	If communication is interrupted after a card-payment request has been sent, the system shall treat the payment result as unresolved and query the Payment Service Provider using the original payment-attempt reference.	Feature-K: Card Payment Recovery and Reconciliation

FR-016	While a card-payment result remains unresolved, the system shall prevent the cashier from initiating another payment, changing to cash, or cancelling the sale.	Feature-K: Card Payment Recovery and Reconciliation
FR-017	If card payment is approved but the system cannot finalize the sale, the system shall retain the payment attempt for recovery and shall not initiate another card charge.	Feature-K: Card Payment Recovery and Reconciliation; Feature-L: Sale Finalization and Durable Recording
FR-018	The system shall automatically reconcile unresolved card-payment attempts with the Payment Service Provider using their original payment-attempt references.	Feature-K: Card Payment Recovery and Reconciliation
FR-019	If reconciliation confirms that a card payment was approved, the system shall finalize the sale without submitting another card charge.	Feature-K: Card Payment Recovery and Reconciliation; Feature-L: Sale Finalization and Durable Recording
FR-020	If reconciliation confirms that a card payment was declined or was not processed, the system shall return the sale to an unpaid state and allow the cashier to retry the card payment, change to cash, or cancel the sale.	Feature-K: Card Payment Recovery and Reconciliation; Feature-H: Payment Method Selection; Feature-D: Sale Lifecycle Management

FR-021	If the Payment Service Provider cannot determine the payment result, the system shall retain the payment attempt in a recovery-pending state and prevent another payment attempt until the result is resolved.	Feature-K: Card Payment Recovery and Reconciliation
FR-022	The system shall allow authorized personnel to view card-payment attempts that remain unresolved or require recovery.	Feature-P: Operational Exception Monitoring; Feature-A: User Authentication and Access Control
FR-023	The system shall record each completed sale with its sale identifier, store, terminal, cashier, date and time, sold items, discount, VAT, final amount, payment method, and applicable payment reference.	Feature-L: Sale Finalization and Durable Recording; Feature-B: Store and Terminal Operating Context
FR-024	The system shall print a receipt using the store receipt printer after a sale is completed successfully.	Feature-M: Receipt Printing and Reprinting
FR-025	If receipt printing fails, the system shall inform the cashier and allow an authorized user to reprint the receipt without reversing or duplicating the completed sale.	Feature-M: Receipt Printing and Reprinting

FR-026	When finalizing a sale, the system shall durably record the completed sale and its pending inventory-update record as one atomic operation.	Feature-L: Sale Finalization and Durable Recording; Feature-N: Inventory Update Outbox
FR-027	If the atomic operation cannot be completed, the system shall not mark the sale as completed. The system shall keep a cash sale open and inform the cashier, or place an already-approved card payment into the payment-recovery process.	Feature-L: Sale Finalization and Durable Recording; Feature-I: Cash Payment Processing; Feature-K: Card Payment Recovery and Reconciliation; Feature-N: Inventory Update Outbox
FR-028	Each pending inventory-update record shall contain a unique sale identifier, store identifier, terminal identifier, transaction date and time, and the product and quantity of each item sold.	Feature-N: Inventory Update Outbox; Feature-B: Store and Terminal Operating Context
FR-029	The system shall asynchronously send each pending inventory-update record to the inventory messaging service.	Feature-O: Inventory Message Delivery
FR-030	If an inventory-update record cannot be delivered, the system shall retain it and retry delivery until the inventory messaging service acknowledges receipt.	Feature-O: Inventory Message Delivery

FR-031	Failure to deliver an already stored inventory-update record shall not reverse or block its corresponding completed sale.	Feature-O: Inventory Message Delivery; Feature-L: Sale Finalization and Durable Recording
FR-032	The system shall allow authorized personnel to view inventory-update records that are awaiting delivery or have repeatedly failed delivery.	Feature-P: Operational Exception Monitoring; Feature-O: Inventory Message Delivery
FR-033	The system shall allow a Manager to view the daily sales summary for the Manager's assigned store and a selected date.	Feature-Q: Daily Sales Reporting; Feature-A: User Authentication and Access Control
FR-034	The system shall allow an Administrator or Head-Office Manager to view the daily sales summary for any selected store.	Feature-Q: Daily Sales Reporting; Feature-A: User Authentication and Access Control
FR-035	The system shall allow an Administrator or Head-Office Manager to view a consolidated daily sales summary across all stores.	Feature-Q: Daily Sales Reporting; Feature-A: User Authentication and Access Control
FR-036	Each daily sales summary shall show the total amount received by cash and the total amount received by credit card separately.	Feature-Q: Daily Sales Reporting
FR-037	The system shall allow an authorized Manager to add a product to the product catalog used by the POS.	Feature-C: Product Catalog and Price Management; Feature-A: User Authentication and Access Control

FR-038	The system shall allow an authorized Manager to update product information and selling prices within the Manager's permitted scope.	Feature-C: Product Catalog and Price Management; Feature-A: User Authentication and Access Control
FR-039	The system shall allow an Administrator to create and deactivate user accounts.	Feature-S: Administration and Configuration Management
FR-040	The system shall allow an Administrator to assign roles and permitted stores to users.	Feature-S: Administration and Configuration Management; Feature-A: User Authentication and Access Control
FR-041	The system shall allow an Administrator to create, update, and deactivate stores.	Feature-S: Administration and Configuration Management; Feature-B: Store and Terminal Operating Context
FR-042	The system shall allow an Administrator to register, activate, and deactivate POS terminals for a store.	Feature-S: Administration and Configuration Management; Feature-B: Store and Terminal Operating Context
FR-043	The system shall allow a cashier to add a product by barcode scan, manual barcode entry, or search by product code or name.	Feature-E: Cart and Sale-Item Management

2. Business Rules

Requirement	Requirement statement	Covered by feature(s)
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BR-001	All product selling prices shall be expressed in Thai baht and shall include VAT according to applicable Thai VAT rules.	Feature-C: Product Catalog and Price Management; Feature-F: VAT-Inclusive Pricing and Total Calculation
BR-002	The sale subtotal shall equal the sum of the extended amounts of all sale items before applying the order-level discount.	Feature-E: Cart and Sale-Item Management; Feature-F: VAT-Inclusive Pricing and Total Calculation
BR-003	A sale may have only one order-level discount. Product-line discounts and multiple combined discounts are not supported.	Feature-G: Order-Level Discount Management
BR-004	When a user enters or changes the Discount Percentage, the system shall calculate the Discount Amount as: $\text{Discount Amount} = \text{Sale Subtotal} \times \text{Discount Percentage} \div 100$.	Feature-G: Order-Level Discount Management
BR-005	A percentage-based Discount Amount shall be rounded to the nearest satang and shall be the amount used to calculate the final sale total.	Feature-G: Order-Level Discount Management; Feature-F: VAT-Inclusive Pricing and Total Calculation
BR-006	When a user enters or changes the Discount Amount directly, the system shall set the Discount Percentage to null and use the entered amount as the discount.	Feature-G: Order-Level Discount Management

BR-007	When the sale contents change while a Discount Percentage is active, the system shall recalculate the Discount Amount using the revised subtotal.	Feature-G: Order-Level Discount Management; Feature-E: Cart and Sale-Item Management
BR-008	The Discount Percentage shall be between 0 and 100 percent, inclusive.	Feature-G: Order-Level Discount Management
BR-009	The Discount Amount shall not be negative and shall not exceed the sale subtotal.	Feature-G: Order-Level Discount Management
BR-010	The final sale amount shall equal the subtotal minus the Discount Amount.	Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-G: Order-Level Discount Management
BR-011	A sale shall use exactly one payment method. Split payment between cash and credit card is not permitted.	Feature-H: Payment Method Selection
BR-012	A cash sale shall not be completed unless the amount tendered is equal to or greater than the final sale amount.	Feature-I: Cash Payment Processing
BR-013	A credit-card sale shall not be completed unless the Payment Service Provider approves payment for the final sale amount.	Feature-J: Card Payment Authorization; Feature-L: Sale Finalization and Durable Recording

BR-014	A sale shall be considered completed after successful payment and permanent recording of the sale. Receipt-printing failure shall not reverse a completed sale.	Feature-L: Sale Finalization and Durable Recording; Feature-M: Receipt Printing and Reprinting
BR-015	A cancelled sale shall not be included in sales totals and shall not produce an inventory message.	Feature-D: Sale Lifecycle Management; Feature-Q: Daily Sales Reporting; Feature-N: Inventory Update Outbox
BR-016	The physical presence of an item at the checkout counter shall be treated as sufficient evidence that it may be sold. The POS shall not synchronously check or block a sale based on the recorded inventory balance.	Feature-E: Cart and Sale-Item Management; Feature-N: Inventory Update Outbox
BR-017	Daily cash and card totals shall include completed sales for the selected store or stores and selected local calendar date, after discounts.	Feature-Q: Daily Sales Reporting
BR-018	A Manager may view daily summaries only for the Manager's assigned store.	Feature-A: User Authentication and Access Control; Feature-Q: Daily Sales Reporting
BR-019	An Administrator or Head-Office Manager may view summaries for any individual store or for all stores combined.	Feature-A: User Authentication and Access Control; Feature-Q: Daily Sales Reporting

BR-020	Returns, refunds, and refund receipts shall not be processed by the POS.	Scope boundary: Verified by the absence of return, refund, and refund-receipt features. No implementation feature is required.
BR-021	The product catalog and selling price of each product shall be global and shall apply to all stores.	Feature-C: Product Catalog and Price Management

3. Non-Functional Requirements

Requirement	Requirement statement	Covered by feature(s)
NFR-001	Under normal operating conditions, the system shall respond to checkout actions, excluding external payment processing and physical printing, within two seconds for at least 95 percent of requests.	Feature-D: Sale Lifecycle Management; Feature-E: Cart and Sale-Item Management; Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-G: Order-Level Discount Management; Feature-H: Payment Method Selection; Feature-I: Cash Payment Processing; Feature-L: Sale Finalization and Durable Recording
NFR-002	A successfully recorded sale shall not be lost because the receipt printer, Payment Service Provider, messaging service, or Inventory Service becomes unavailable afterward.	Feature-L: Sale Finalization and Durable Recording; Feature-M: Receipt Printing and Reprinting; Feature-J: Card Payment Authorization; Feature-K: Card Payment Recovery and Reconciliation; Feature-N: Inventory Update Outbox; Feature-O: Inventory Message Delivery

NFR-003	Inventory messages shall be stored durably and delivered using a mechanism that supports retry without losing completed sales.	Feature-N: Inventory Update Outbox; Feature-O: Inventory Message Delivery
NFR-004	Each inventory message shall contain a unique transaction identifier so that repeated delivery does not cause the Inventory Service to deduct the same stock more than once.	Feature-N: Inventory Update Outbox; Feature-O: Inventory Message Delivery
NFR-005	Failure of the Payment Service Provider shall not prevent the POS from completing an otherwise valid cash sale.	Feature-H: Payment Method Selection; Feature-I: Cash Payment Processing; Feature-J: Card Payment Authorization
NFR-006	The system shall enforce role-based access to checkout, reporting, and product-management functions.	Feature-A: User Authentication and Access Control

NFR-007	The POS shall not store sensitive card details and shall communicate with the card machine and Payment Service Provider through secure, approved interfaces.	Feature-J: Card Payment Authorization; Feature-K: Card Payment Recovery and Reconciliation
NFR-008	The system shall record the responsible user, date, time, store, and terminal for completed sales, sale cancellations, discounts, and product or price changes.	Feature-R: Audit Trail
NFR-009	The system shall support multiple stores and multiple concurrently operating POS terminals without transaction-identifier collisions.	Feature-B: Store and Terminal Operating Context; Feature-L: Sale Finalization and Durable Recording
NFR-010	The system shall represent monetary values in Thai baht to two decimal places and use the store's local date and time for daily reporting.	Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-B: Store and Terminal Operating Context; Feature-Q: Daily Sales Reporting

NFR-011	The system shall integrate with the approved receipt printer, card machine, Payment Service Provider, and asynchronous inventory-messaging interface.	Feature-M: Receipt Printing and Reprinting; Feature-J: Card Payment Authorization; Feature-O: Inventory Message Delivery
NFR-012	A terminal shall require connectivity to the Kaching backend to start or complete a sale; offline sales are not supported.	Feature-D: Sale Lifecycle Management; System constraint: Online connectivity required
NFR-013	The system shall support 20 stores, 10 active terminals per store, 200 concurrent users, and at least 20 completed sales per second.	Cross-cutting: Capacity, scalability, concurrency, and performance
NFR-014	The Kaching backend shall provide 99.5 percent monthly availability.	Cross-cutting: Availability and reliability
NFR-015	The production service shall meet an RPO of 15 minutes and an RTO of 4 hours.	Cross-cutting: Backup and disaster recovery
NFR-016	The English-only user interface shall support desktop and touchscreen use at 1280 x 720 or higher and meet WCAG 2.1 AA color contrast.	Cross-cutting UI: Usability, compatibility, and accessibility